Mid-Term Examination for Thermodynamics and Statistical Mechanics 2025-2026(2)

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Problem 1 A system consists of N **distinguishable, independent particles** obeying **Boltzmann statistics**. Each particle can occupy one of three non-degenerate energy levels: $-E$, 0, and E. The system is in thermal contact with a heat reservoir at temperature T.

Define $\beta = \frac{1}{k_B T}$. The single-particle partition function is:

$$z = e^{\beta E} + 1 + e^{-\beta E} = 1 + 2\cosh(\beta E)$$

The occupation probabilities for the three levels are:

$$P(-E) = \frac{e^{\beta E}}{z}, \quad P(0) = \frac{1}{z}, \quad P(E) = \frac{e^{-\beta E}}{z}$$

(a) Find the entropy at $T = 0$ K.

(b) Find the maximum entropy.

(c) Find the minimum energy of the system.

(d) Find the total partition function Z for the N-particle system.

(e) Find the average energy $\langle U \rangle$ of the system.

(f) Evaluate the integral

$$\int_0^{\infty} \frac{C(T)}{T} \mathrm{d}T$$

.

Solution.

(a) Entropy at $T = 0$ K

As $T \to 0$, we have $\beta \to \infty$, so $P(-E) \to 1$ and $P(0), P(E) \to 0$.

Every particle occupies the ground state $-E$. There is only **one microstate** (all N particles on $-E$), so:

$$S = k_B \ln 1 = \boxed{0}$$

This is consistent with the **Third Law of Thermodynamics**.

(b) Maximum Entropy

Maximum entropy corresponds to $T \to \infty$ ($\beta \to 0$), where all three levels are **equally probable**.

For N distinguishable particles, a configuration with occupation numbers (n_1, n_2, n_3) where $n_1 + n_2 + n_3 = N$ has a multiplicity:

$$W = \frac{N!}{n_1!\, n_2!\, n_3!}$$

W is maximized when $n_1 = n_2 = n_3 = \frac{N}{3}$. Using **Stirling's approximation** ($\ln n! \approx n \ln n - n$):

$$
\begin{aligned}
S_{\max} &= k_B \ln\left(\frac{N!}{\frac{N}{3}!^3}\right) \\
&\approx k_B\left[N\ln N - 3\cdot\frac{N}{3}\cdot\ln\left(\frac{N}{3}\right)\right] \\
&= k_B N \ln 3
\end{aligned}
$$

$$\boxed{S_{\max} = \mathrm{N}\ k_B\ \ln\ 3}$$

(c) Minimum Energy

The minimum energy occurs when **all** N particles are in the lowest level $-E$:

$$\boxed{U_{\min} = -\mathrm{N\ E}}$$

(d) Partition Function

Since the N particles are **distinguishable and independent**, the total partition function is:

$$\boxed{\mathrm{Z} = z^N = \left[1 + 2\cosh\left(\frac{E}{k_B T}\right)\right]^N}$$

(e) Average Energy

The average energy of a single particle:

$$
\begin{aligned}
\langle\varepsilon\rangle &= (-E)\cdot\frac{e^{\beta E}}{z} + (0)\cdot\frac{1}{z} + (E)\cdot\frac{e^{-\beta E}}{z} \\
&= E\frac{e^{-\beta E} - e^{\beta E}}{z} \\
&= -\frac{2E\sinh(\beta E)}{1 + 2\cosh(\beta E)}
\end{aligned}
$$

For N particles:

$$\boxed{\langle\ \mathrm{U}\ \rangle = -\frac{2NE\sinh\left(\frac{E}{k_B T}\right)}{1+2\cosh\left(\frac{E}{k_B T}\right)}}$$

(f) Integral of $\frac{C(T)}{T}$

We need to evaluate:

$$\int_0^\infty \frac{C(T)}{T}\,\mathrm{d}T = ?$$

Since the energy levels are fixed (no volume degree of freedom), entropy depends only on temperature:

$$\mathrm{d}S = \frac{C(T)}{T}\,\mathrm{d}T$$

Therefore, the integral equals the **total entropy change**:

$$\int_0^\infty \frac{C(T)}{T}\,\mathrm{d}T = \int_0^\infty \mathrm{d}S = S(T=\infty) - S(T=0)$$

Substituting results from parts (a) and (b):

$$\boxed{\int_0^\infty \frac{C(T)}{T}\,\mathrm{d\ T} = \mathrm{N}\ k_B\ \ln\ 3}$$

Problem 2 Consider a collection of N two-level systems in thermal equilibrium at temperature T. Each system has only two energy levels: a ground state with energy 0 and an excited state with energy ε. Find and plot vs. temperature:

(a) The probability that a subsystem is in the excited state

(b) The entropy of the entire collection

Solution.

(a) Probability of Excited State

For a single two-level system, the partition function is:

$$z = 1 + e^{-\beta\varepsilon}, \quad \beta = \frac{1}{k_B T}$$

By the Boltzmann distribution, the probability of occupying the excited state is:

$$\boxed{P_{\text{ex}} = \frac{e^{-\beta\varepsilon}}{1+e^{-\beta\varepsilon}} = \frac{1}{e^{\frac{\varepsilon}{k_B T}}+1}}$$

Physical Checks

T	P_{ex}	Interpretation
$T \to 0$	$P_{\text{ex}} \to 0$	All systems frozen in ground state
$T \to \infty$	$P_{\text{ex}} \to \frac{1}{2}$	Both levels equally populated

Temperature Dependence (Sketch)

P_{ex} vs T sketch:

```
P_ex
1/2 | · · · · · · · · · · ·  (asymptote)
 |      · · · · ·
 |    · · ·
 |   · ·
 |  · ·
 | ·
 | ·
 |·
 +----------------------- T
   kT ~ ε
```

The curve rises smoothly from 0, has the steepest slope near $k_B T \sim \varepsilon$, and asymptotically approaches $\frac{1}{2}$.

—

(b) Entropy of the Entire Collection

Derivation

Method 1: Direct from the Gibbs (Shannon) entropy formula

Define $p = P_{\text{ex}}$ and $q = 1 - p = P_{\text{ground}} = \frac{1}{1+e^{-\beta\varepsilon}}$.

Since the N systems are independent, the total entropy is:

$$\boxed{\mathrm{S} = -\,\mathrm{N}\; k_B\,[p\ln p + (1-p)\ln(1-p)]}$$

where $p = \frac{1}{e^{\frac{\varepsilon}{k_B T}}+1}$.

Method 2: From the partition function (more formal)

The total partition function for N distinguishable subsystems:

$$Z = z^N = \left(1 + e^{-\beta\varepsilon}\right)^N$$

The Helmholtz free energy:

$$F = -k_B T \ln Z = -N k_B T \ln\left(1 + e^{-\frac{\varepsilon}{k_B T}}\right)$$

The entropy:

$$S = -\left(\partial \frac{F}{\partial} T\right)_V$$

Computing the derivative:

$$\frac{\partial(F)}{\partial(T)} = -N k_B \ln\left(1 + e^{-\beta\varepsilon}\right) - N k_B T \cdot \frac{e^{-\beta\varepsilon} \cdot \left(\frac{\varepsilon}{k_B} T^2\right)}{1 + e^{-\beta\varepsilon}}$$

Therefore:

$$\boxed{\mathrm{S} = \mathrm{N}\ k_B\ \ln\left(1 + e^{-\frac{\varepsilon}{k_B T}}\right) + \frac{N\varepsilon}{T} \cdot \frac{e^{-\frac{\varepsilon}{k_B T}}}{1+e^{-\frac{\varepsilon}{k_B T}}}}$$

"**Equivalence check:** Expanding Method 2′s result with p and q:

$S = -N k_B [p \ln p + q \ln q]$ — exactly Method 1.“

Temperature Dependence (Sketch)

S vs ***T*** sketch:

```
S
Nkln2 | · · · · · · · · · · ·  (asymptote)
  |   · · · · ·
  |  · · ·
  |  · ·
  | · ·
  | ·
  | ·
  |·
  +------------------------ T
   kT ~ ε
```

The entropy rises from 0 at $T = 0$ and asymptotically approaches $N k_B \ln 2$ at high temperature.

T	S	Interpretation
$T \to 0$	$S \to 0$	Perfect order: all in ground state
$T \to \infty$	$S \to N k_B \ln 2$	Maximum disorder: equal occupation of 2 levels

"**Note on the maximum:** The N subsystems each have 2 states, giving at most 2^N microstates, so $S_{\max} = k_B \ln 2^N = N k_B \ln 2$. This maximum is achieved at infinite temperature when both levels are equally probable."

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Problem 3

By using Maxwell relations, prove:

$$\left(\partial \frac{C_p}{\partial} p\right)_T = -T \left(\partial^2 \frac{V}{\partial} T^2\right)_p$$

Given hint:

$$\left(\partial\frac{S}{\partial}p\right)_T = -\left(\partial\frac{V}{\partial}T\right)_p$$

Solution.

Step 1 — Definition of C_p
Start from the definition of the heat capacity at constant pressure:

$$C_p = T\left(\partial\frac{S}{\partial}T\right)_p$$

Step 2 — Differentiate both sides with respect to p (at constant T)

$$\left(\partial\frac{C_p}{\partial}p\right)_T = \left(\frac{\partial}{\partial}p\left[T\left(\partial\frac{S}{\partial}T\right)_p\right]\right)_T$$

Since T is held constant in the outer derivative, it passes through:

$$= T\left(\frac{\partial}{\partial}p\left(\partial\frac{S}{\partial}T\right)_p\right)_T$$

Step 3 — Swap the order of differentiation
Because S is a *state function*, its second partial derivatives are equal (Clairaut's theorem):

$$\left(\frac{\partial}{\partial}p\partial\frac{S}{\partial}T\right)_T = \left(\frac{\partial}{\partial}T\partial\frac{S}{\partial}p\right)_p$$

So:

$$\left(\partial\frac{C_p}{\partial}p\right)_T = T\left(\frac{\partial}{\partial}T\left(\partial\frac{S}{\partial}p\right)_T\right)_p$$

Step 4 — Apply the Maxwell relation
From the *Gibbs free energy* $G = H - TS$, with $\mathrm{d}G = -S\,\mathrm{d}T + V\,\mathrm{d}p$, we get the Maxwell relation:

$$\left(\partial\frac{S}{\partial}p\right)_T = -\left(\partial\frac{V}{\partial}T\right)_p$$

Substitute this in:

$$\left(\partial\frac{C_p}{\partial}p\right)_T = T\left(\frac{\partial}{\partial}T\left[-\left(\partial\frac{V}{\partial}T\right)_p\right]\right)_p$$

Step 5 — Simplify

$$\boxed{\left(\partial\tfrac{C_p}{\partial}p\right)_T = -\,\mathrm{T}\,\left(\partial^2\tfrac{V}{\partial}T^2\right)_p} \quad \blacksquare$$

—

Summary of Logic Chain

$$C_p = T\left(\partial\frac{S}{\partial}T\right)_p \to \text{diff/diff p swap order} \to \text{Maxwell} - T\left(\partial^2\frac{V}{\partial}T^2\right)_p$$

"**Physical meaning:** This identity connects the pressure dependence of heat capacity to the *non-linearity* of the equation of state (V vs T). If V is exactly linear in T (e.g., ideal gas), the right side vanishes and C_p is independent of pressure — which is indeed the case for an ideal gas."

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Problem 4

Problem 4: Average Energy in the Classical Canonical Ensemble

A particle obeys the classical canonical distribution with energy:

$$\varepsilon = \frac{1}{2m}(p_x^2 + p_y^2 + p_z^2) + ax^2 + bx$$

where a and b are constants. Find the average energy of the particle.

Solution.

Step 1 — Partition Function

For a classical canonical ensemble:

$$z = \int e^{-\beta\varepsilon} \mathrm{d}^3 p \mathrm{d}^3 \frac{x}{h^3}, \quad \beta = \frac{1}{k_B T}$$

Since ε separates into independent momentum and position parts, the integral *factorizes*:

$$z \simeq z_{\text{kinetic}} \cdot z_{\text{potential}}$$

Step 2 — Momentum Part (Kinetic Energy)

Each momentum component gives a Gaussian integral:

$$z_{p_i} = \int_{-\infty}^{\infty} e^{-\beta\frac{p_i^2}{2m}} \, \mathrm{d}p_i = \sqrt{\frac{2\pi m}{\beta}}$$

Contribution from all three components:

$$\ln z_{\text{kinetic}} = \frac{3}{2} \ln \frac{2\pi m}{\beta}$$

By the *equipartition theorem*, each quadratic momentum term contributes $\frac{1}{2}k_B T$:

$$\langle \varepsilon_k \rangle = \frac{3}{2} k_B T$$

Step 3 — Position Part (Potential Energy)

$$z_x = \int_{-\infty}^{\infty} e^{-\beta(ax^2+bx)} \, \mathrm{d}x$$

Complete the square in the exponent:

$$ax^2 + bx = a\left(x + \frac{b}{2a}\right)^2 - \frac{b^2}{4a}$$

Therefore:

$$z_x = e^{\beta\frac{b^2}{4a}} \int_{-\infty}^{\infty} e^{-\beta a\left(x+\frac{b}{2a}\right)^2} \, \mathrm{d}x = e^{\beta\frac{b^2}{4a}} \sqrt{\frac{\pi}{\beta a}}$$

Step 4 — Compute the Average Energy

$$\langle \varepsilon \rangle = -\frac{\partial \ln z}{\partial \beta}$$

Combining all parts (factors of h, volume integrals over y, z cancel):

$$\ln z = -\frac{3}{2} \ln \beta + \frac{\beta b^2}{4a} - \frac{1}{2} \ln \beta + \text{const.} = -2 \ln \beta + \frac{\beta b^2}{4a} + \text{const.}$$

Differentiating:

$$\langle \varepsilon \rangle = -\frac{\partial}{\partial \beta}\left(-2\ln\beta + \frac{\beta b^2}{4a}\right) = \frac{2}{\beta} - \frac{b^2}{4a}$$

$$\boxed{\langle\ \varepsilon\ \rangle = 2\ k_B\ \mathrm{T} - \frac{b^2}{4a}}$$

Verification via Equipartition Theorem

We can decompose the result term by term:

Term	Contribution	Method
$\frac{p_x^2}{2m}$	$\frac{1}{2}k_BT$	Equipartition (quadratic)
$\frac{p_y^2}{2m}$	$\frac{1}{2}k_BT$	Equipartition (quadratic)
$\frac{p_z^2}{2m}$	$\frac{1}{2}k_BT$	Equipartition (quadratic)
ax^2	$\frac{1}{2}k_BT$	Equipartition (quadratic)
bx	$-\frac{b^2}{2a}$	Linear term: shift of origin
Cross-term corr.	$+\frac{b^2}{4a}$	From completing the square
Total	$2k_BT - \frac{b^2}{4a}$	

Physical Interpretation

The potential $U(x) = ax^2 + bx$ has its minimum at $x_0 = -\frac{b}{2a}$, with:

$$U_{\min} = -\frac{b^2}{4a}$$

So the result reads:

$$\langle \varepsilon \rangle = \underbrace{\frac{3}{2}k_BT}_{\text{3D kinetic}} + \underbrace{\frac{1}{2}k_BT}_{\text{oscillator}} + \underbrace{-\frac{b^2}{4a}}_{U_{\min}}$$

The linear term bx simply *shifts the equilibrium position* to $x_0 = -\frac{b}{2a}$ without changing the oscillation frequency — it only contributes the constant $U_{\min}$ to the average energy.

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Problem 5

Problem 5: Fraction of Molecules Below the Most Probable Speed

Suppose the number of molecules in a classical ideal gas whose speed is less than the most probable speed v_p is N_p, and the total number of gas molecules is N. Prove that the ratio $r \stackrel{\text{def}}{=} \frac{N_p}{N}$ is *independent of the temperature* of the gas.

Solution.

Step 1 — Recall the Maxwell—Boltzmann Distribution

The speed distribution of a classical ideal gas is:

$$f(v) = 4\pi \frac{m}{2\pi k_BT}^{\frac{3}{2}} v^2 \exp\left(-\frac{mv^2}{2k_BT}\right)$$

The *most probable speed* (found by maximizing $f(v)$) is:

$$v_p = \sqrt{\frac{2k_BT}{m}}$$

Step 2 — Write the Ratio r

$$r = \frac{N_p}{N} = \int_0^{v_p} f(v)\,\mathrm{d}v$$

Step 3 — Change of Variable: Let $u = \frac{v}{v_p}$

This is the *key substitution*. Set:

$$v = uv_p = u\sqrt{\frac{2k_BT}{m}}, \quad \mathrm{d}v = v_p\,\mathrm{d}u$$

When $v = 0 \to u = 0$, and $v = v_p \to u = 1$.

Substituting into the exponent:

$$\frac{mv^2}{2k_BT} = \frac{m}{2k_BT} \cdot u^2 \cdot \frac{2k_BT}{m} = u^2$$

Substituting into the prefactor:

$$4\pi\frac{m}{2\pi k_BT}^{\frac{3}{2}} v^2 = 4\pi\frac{m}{2\pi k_BT}^{\frac{3}{2}} u^2 \cdot \frac{2k_BT}{m}$$

Multiplying by $\mathrm{d}v = v_p\,\mathrm{d}u = \sqrt{\frac{2k_BT}{m}}\mathrm{d}u$:

$$f(v)\,\mathrm{d}v = 4\pi\frac{m}{2\pi k_BT}^{\frac{3}{2}} \cdot \frac{2k_BT}{m} \cdot \sqrt{\frac{2k_BT}{m}} \cdot u^2 e^{-u^2}\,\mathrm{d}u$$

The T-dependent prefactors simplify:

$$4\pi \cdot \frac{m^{\frac{3}{2}}}{(2\pi k_BT)^{\frac{3}{2}}} \cdot \frac{(2k_BT)^{\frac{3}{2}}}{m^{\frac{3}{2}}} = \frac{4\pi}{(2\pi)^{\frac{3}{2}}} \cdot 2^{\frac{3}{2}} = \frac{4}{\sqrt{\pi}}$$

Step 4 — The Result

$$\boxed{\mathrm{r} = \frac{N_p}{N} = \frac{4}{\sqrt{\pi}} \int_0^1 u^2\, e^{-u^2}\,\mathrm{d}\,\mathrm{u}}$$

Every factor in this expression is a pure number — there is no T anywhere. The upper limit is 1 (not a function of T), and the integrand $u^2e^{-u^2}$ contains no T.

Therefore, r is *independent of temperature*. ■

Step 5 — Numerical Value

Evaluate the integral by parts ($w = u$, $\mathrm{d}w = ue^{-u^2}\,\mathrm{d}u$):

$$\int_0^1 u^2e^{-u^2}\,\mathrm{d}u = \left[-\frac{u}{2}e^{-u^2}\right]_0^1 + \frac{1}{2}\int_0^1 e^{-u^2}\,\mathrm{d}u = -\frac{1}{2e} + \frac{\sqrt{\pi}}{4}\,\mathrm{erf}(1)$$

$$\approx -0.1839 + 0.4431 \cdot 0.8427 \approx 0.1895$$

Therefore:

$$r = \frac{4}{\sqrt{\pi}} \cdot 0.1895 \approx 2.257 \cdot 0.1895$$

$$\boxed{\mathrm{r} \approx 0.428 \quad \text{(i.e., about 42.8\%)}}$$

Physical Intuition

Why is this ratio temperature-independent?

"At higher T, the distribution *broadens* and shifts to larger speeds — but v_p itself shifts to a larger value by exactly the same factor. The shape of the distribution *relative to* v_p is universal: the substitution $u = \frac{v}{v_p}$ removes all T-dependence. This is a self-similarity property of the Maxwell—Boltzmann distribution — it always "looks the same" when measured in units of its own most probable speed."

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Problem 6

Problem 6: Fluctuations in Internal Energy and Heat Capacity

Derive that the fluctuations in internal energy at thermal equilibrium are proportional to the heat capacity, and explain why these fluctuations can be neglected under such conditions.

Solution.

Setup: Canonical Ensemble

A system with fixed N particles and volume V is in thermal equilibrium with a heat bath at temperature T. The internal energy is not strictly constant — it fluctuates around a mean value $\langle E\rangle$. We use the canonical ensemble with $\beta = \frac{1}{k_B T}$.

The partition function and probability of energy state r are:

$$Z = \sum_r e^{-\beta E_r}, \quad P_r = \frac{e^{-\beta E_r}}{Z}$$

Part 1: Deriving the Fluctuation—Heat Capacity Relation

— Step 1 — Mean Energy

$$\langle E\rangle = \sum_r E_r P_r = \frac{1}{Z}\sum_r E_r e^{-\beta E_r} = -\frac{\partial \ln Z}{\partial \beta}$$

— Step 2 — Mean Square Energy

$$\langle E^2\rangle = \frac{1}{Z}\sum_r E_r^2 e^{-\beta E_r} = \frac{1}{Z}\frac{\partial^2 Z}{\partial \beta^2}$$

— Step 3 — Variance of Energy

The variance is:

$$\sigma_E^2 = \langle E^2\rangle - \langle E\rangle^2$$

We can write this compactly using the derivative identity:

$$\sigma_E^2 = \langle E^2\rangle - \langle E\rangle^2 = \frac{\partial^2 \ln Z}{\partial \beta^2}$$

Proof:

$$\frac{\partial \ln Z}{\partial \beta} = \frac{1}{Z}\frac{\partial Z}{\partial \beta} = -\langle E\rangle$$

Differentiating again:

$$\frac{\partial^2 \ln Z}{\partial \beta^2} = \frac{\partial}{\partial \beta}(-\langle E\rangle) = -\frac{\partial \langle E\rangle}{\partial \beta}$$

But we also need to show this equals $\langle E^2\rangle - \langle E\rangle^2$. Let us compute directly:

$$\sigma_E^2 = \langle (E - \langle E\rangle)^2\rangle = \langle E^2\rangle - \langle E\rangle^2$$

Using $e^{-\beta E_r}$:

$$\sigma_E^2 = \sum_r E_r^2 P_r - \left(\sum_r E_r P_r\right)^2 = \frac{\partial^2 \ln Z}{\partial \beta^2}$$

This is the standard result:

$$\boxed{\sigma_E^2 = \langle\ E^2\ \rangle - \langle\ \mathrm{E}\ \rangle^2 = \frac{\partial^2 \ln Z}{\partial \beta^2}}$$

— Step 4 — Connect to Heat Capacity

The constant-volume heat capacity is:

$$C_V = \left(\partial \langle E \frac{\rangle}{\partial} T\right)_{N,V}$$

Using the chain rule $\frac{\partial}{\partial T} = \frac{\partial \beta}{\partial T}\frac{\partial}{\partial \beta} = -\frac{1}{k_B T^2}\frac{\partial}{\partial \beta}$:

$$C_V = -\frac{1}{k_B T^2}\frac{\partial \langle E\rangle}{\partial \beta} = -\frac{1}{k_B T^2}\left(-\sigma_E^2\right) = \frac{\sigma_E^2}{k_B T^2}$$

Rearranging:

$$\boxed{\sigma_E^2 = k_B\ T^2\ C_V}$$

This is the *fluctuation—dissipation relation* for energy: the mean-square energy fluctuation is directly proportional to the heat capacity.

Part 2: Why Fluctuations Are Negligible for Macroscopic Systems

— Relative Fluctuation

The physically meaningful quantity is the *relative* fluctuation:

$$\frac{\sigma_E}{\langle} E\rangle = \frac{\sqrt{k_B T^2 C_V}}{\langle} E\rangle$$

For a macroscopic system with $N \sim 10^{23}$ particles:

- $\langle E\rangle \sim N$ (extensive)
- $C_V \sim N$ (extensive)

Therefore:

$$\frac{\sigma_E}{\langle} E\rangle \sim \frac{\sqrt{N}}{N} = \frac{1}{\sqrt{N}}$$

— Numerical Estimate

$$\frac{\sigma_E}{\langle} E\rangle \sim \frac{1}{\sqrt{10^{23}}} \sim 10^{-11.5}$$

This is an astonishingly small fraction — the energy fluctuates by about *one part in 100 billion.*

— Physical Explanation

Aspect	Reason
Statistical averaging	With $N \sim 10^{23}$ particles, the central limit theorem ensures fluctuations scale as $\sqrt{N}$ while the mean scales as N
Thermal bath coupling	The heat reservoir is macroscopic; even large energy transfers between system and bath are negligible relative to the total energy of either

Measurement resolution	No macroscopic instrument can detect a 10^{-11} relative fluctuation

— When Fluctuations Matter

Fluctuations become significant only when:

$$N \text{ is small} \longrightarrow \frac{\sigma_E}{\langle} E\rangle \sim \frac{1}{\sqrt{N}} \sim 1$$

This occurs in *nanosystems*, *mesoscopic physics*, or *critical phenomena* (where the effective number of correlated degrees of freedom diverges).

Summary

$$\boxed{\sigma_E^2 = k_B\, T^2\, C_V \Rightarrow \frac{\sigma_E}{\langle}\ \mathrm{E}\ \rangle \sim \frac{1}{\sqrt{N}}}$$

"Energy fluctuations are proportional to heat capacity. For macroscopic systems ($N \sim 10^{23}$), the relative fluctuation is of order 10^{-12}, making the thermodynamic energy effectively a *sharp, deterministic quantity* — which is why the canonical ensemble (fixed T, fluctuating E) gives identical predictions to the microcanonical ensemble (fixed E) for macroscopic systems."

Summary of Results

Problem	Part	Quantity	Result
1	(a)	Entropy at $T = 0$	0
1	(b)	Maximum entropy	$Nk_B \ln 3$
1	(c)	Minimum energy	$-NE$
1	(d)	Partition function	$\left[1 + 2\cosh\left(\frac{E}{k_B}T\right)\right]^N$
1	(e)	Average energy	$-2NE\frac{\sinh\left(\frac{E}{k_B}T\right)}{\left[1+2\cosh\left(\frac{E}{k_B}T\right)\right]}$
1	(f)	Integral of $\frac{C(T)}{T}$	$Nk_B \ln 3$
2	(a)	Excited state prob.	$\frac{1}{e^{\frac{\epsilon}{k_B}T}+1}$
2	(b)	Entropy	$-Nk_B[p\ln p + (1-p)\ln(1-p)]$
3		Maxwell relation proof	$\left(\partial\frac{C_p}{\partial}p\right)_T = -T\left(\partial^2\frac{V}{\partial}T^2\right)_p$
4		Average energy	$2k_BT - \frac{b^2}{4a}$
5		Ratio $\frac{N_p}{N}$	≈ 0.428
6		Energy fluctuations	$\sigma_E^2 = k_BT^2C_V$